PROJECT AND TEAM INFORMATION

Project Title

Code Shield : Source Code Similarity and Plagiarism Detection System

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Many students submit programming assignments, so checking them manually for similar or copied code can be difficult and time-consuming. Simple text comparison may not work properly because students can change variable names, comments, spacing, or formatting while keeping the same logic. Code Shield is designed to detect such similarities in C/C++ programs. It uses ideas from MOSS and the Winnowing technique. The code is first processed and normalized, then divided into small parts, hashed, and compared using fingerprints. The main aim is to help teachers quickly identify potentially similar submissions.

State of the Art / Current solution

Currently, source code similarity is detected using tools and techniques such as MOSS and JPlag. These systems compare programs using tokens, code structure, and fingerprinting methods instead of only comparing the text. They can detect similarities even when variable names, comments, or formatting are changed. However, these tools can be complex for a small academic project and may not provide a simple, customizable solution for specific C/C++ assignments. Code Shield aims to provide a simpler prototype using code normalization, k-grams, hashing, and Winnowing-based fingerprints.

Project Goals and Milestones

The main goal is to develop a C/C++ source-code similarity detection system that can compare multiple submissions and highlight suspicious matching regions. Initial milestones are:

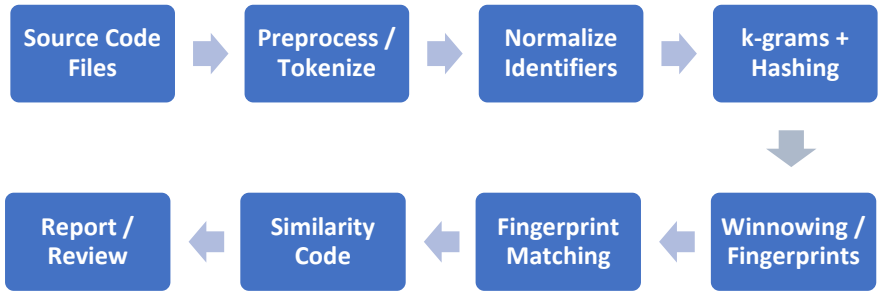
1. Read and collect source-code files.
2. Remove comments and unnecessary formatting from the code.
3. Tokenize the source code into relevant tokens.
4. Normalize identifiers so that simple variable renaming does not completely hide similarity.
5. Generate k-grams from the normalized token sequence.
6. Calculate hash values for the generated k-grams.
7. Apply the Winnowing algorithm to select important fingerprints.
8. Store fingerprints along with file and position information.
9. Compare fingerprints and calculate a similarity score.
10. Generate a simple report showing similar files and matching code locations.

A further milestone is to handle common instructor-provided or base code so that legitimate shared code does not unnecessarily increase the similarity.

Project Approach

The project will be developed using C/C++. First, the submitted source code files will be read and basic things like comments and extra spaces will be removed. After that, the code will be divided into tokens such as keywords, variables and operators. Variable names will be normalized so that changing variable names does not easily avoid detection. Then, the tokens will be divided into small groups called k-grams and hash values will be generated for them. The Winnowing method will be used to select important fingerprints from these hash values. The fingerprints will be stored along with their file and position information. Finally, fingerprints from different student files will be compared to find similar code and calculate a similarity score. The system will then show the similar files and the parts of code where matching is found.

System Architecture



Project Outcome / Deliverables

The main outcome of this project will be a working Code Shield system that can compare C/C++ programs and find similar code between different student submissions. It will remove unnecessary differences like comments, extra spaces and different variable names before comparing the code. The system will use tokenization, k-grams, hashing and the Winnowing method to find important parts of the code and compare them. It will also show the similarity percentage between two programs and highlight the code sections that are similar. The final system will generate a simple report which can help teachers identify submissions that may need further checking.

Assumptions

The project will mainly focus on C/C++ source code. The submitted files should be in a readable source-code format. The system will compare the code after removing comments, extra spaces and other unnecessary differences. Common code given by the teacher can be kept separate so that it does not affect the similarity result. The similarity percentage will only show that two programs are similar; the final decision about plagiarism will be made by the teacher.

References

1. S. Schleimer, D. S. Wilkerson and A. Aiken, "Winnowing: Local Algorithms for Document Fingerprinting," ACM SIGMOD, 2003.
2. Stanford University, "MOSS: A System for Detecting Software Similarity."
3. Stanford University, Alex Aiken Publications, "Winnowing: Local Algorithms for Document Fingerprinting."